Abstract and Virtualization

- When you use cloud computing, you are accessing pooled resources using a technique called virtualization. Virtualization assigns a logical name for a physical resource and then provides a pointer to that physical resource when a request is made.

- Virtualization provides a means to manage resources efficiently because the mapping of virtual resources to physical resources can be both dynamic and facile.

- Virtualization is dynamic in that the mapping can be assigned based on rapidly changing conditions, and it is facile because changes to a mapping assignment can be nearly instantaneous.

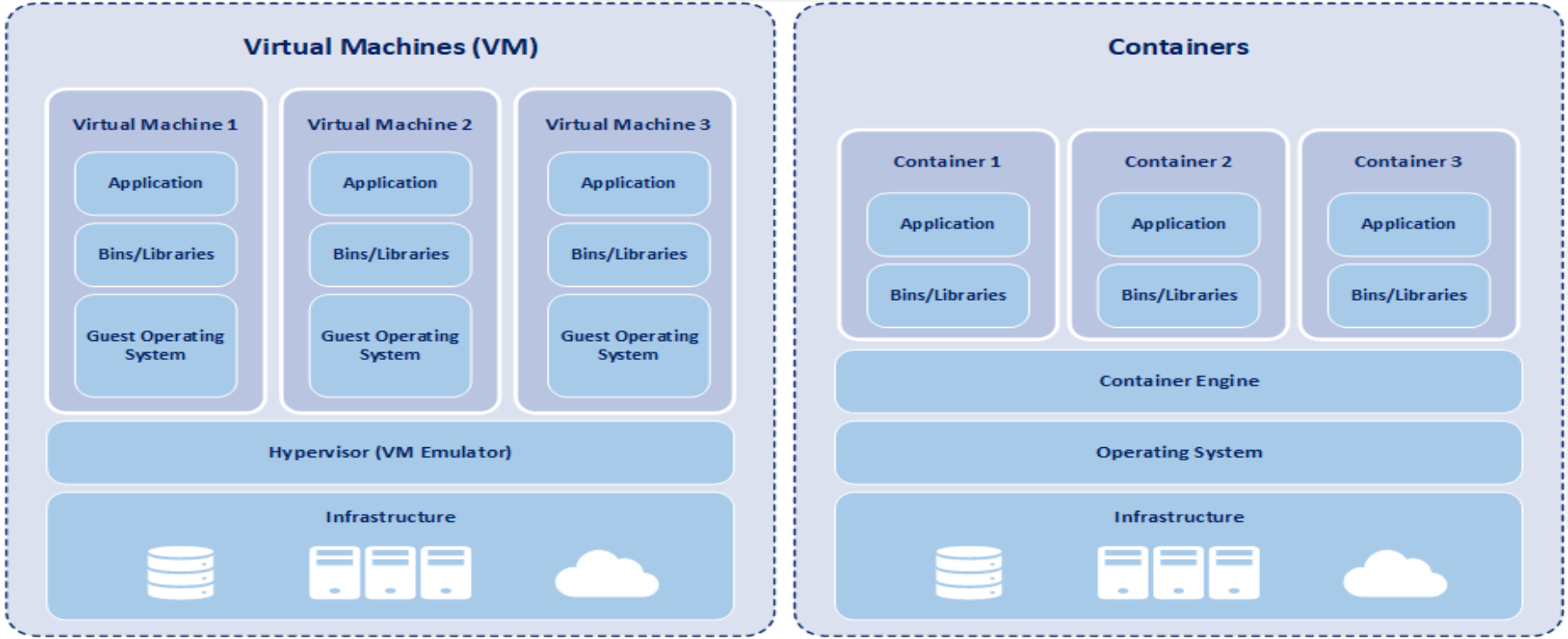
These are among the different types of virtualization that are characteristic of cloud computing:

- **Access:** A client can request access to a cloud service from any location.

- **Application:** A cloud has multiple application instances and directs requests to an instance based on conditions.

- **CPU:** Computers can be partitioned into a set of virtual machines with each machine being assigned a workload. Alternatively, systems can be virtualized through load-balancing technologies.
- **Storage:** Data is stored across storage devices and often replicated for redundancy.

Abstract and Virtualization



Abstract and Virtualization

Virtualization Type	Key Benefit	Example Use Case
• Server Virtualization	• Maximizes resource utilization	• Cloud hosting & data centers
• Storage Virtualization	• Better storage efficiency & management	• Enterprise storage & disaster recovery
• Network Virtualization	• Improves network performance & security	• Cloud networking & data centers
• Application Virtualization	• Eliminates OS dependency	• Running legacy & SaaS applications
• Desktop Virtualization	• Enables remote access & flexibility	• VDI (Virtual Desktop Infrastructure) for remote work
• Data Virtualization	• Unifies data from multiple sources	• Business intelligence & big data analytics
• GPU Virtualization	• Enhances graphical & AI processing power	• AI training, gaming, & video rendering

Abstract and Virtualization

- To enable these characteristics, resources must be highly configurable and flexible. You can define the features in software and hardware that enable this flexibility as conforming to one or more of the following mobility patterns:
- P2V: Physical to Virtual
- V2V: Virtual to Virtual
- V2P: Virtual to Physical
- P2P: Physical to Physical
- D2C: Datacenter to Cloud
- C2C: Cloud to Cloud
- C2D: Cloud to Datacenter
- D2D: Datacenter to Datacenter

Abstract and Virtualization

- According to Gartner, virtualization is a key enabler of the **first four of five key attributes of cloud computing:**
- **Service-based:** A service-based architecture is where clients are abstracted from service providers through service interfaces.
- **Scalable and elastic:** Services can be altered to affect capacity and performance on demand.
- **Shared services:** Resources are pooled in order to create greater efficiencies.
- **Metered usage:** Services are billed on a usage basis.
- **Internet delivery:** The services provided by cloud computing are based on Internet protocols and formats.

Levels of Virtualization

- Virtualization in cloud computing operates across multiple layers, each optimizing resource utilization and system performance. The common virtualization layers include:
- **Instruction Set Architecture (ISA) Level** – Virtualization occurs at the processor level, enabling different operating systems to run on a single CPU through emulation. Example: Binary translation in x86 architecture.
- **Hardware Level** – Uses a hypervisor to abstract physical hardware, allowing multiple virtual machines (VMs) to run on a single system. Examples: VMware ESXi, Microsoft Hyper-V.
- **Operating System Level** – Also known as containerization, this enables multiple isolated user-space environments to share the same OS kernel. Examples: Docker, LXC.
- **Library Support Level** – Virtualization is achieved by modifying standard libraries so applications can run on different platforms without requiring OS modifications. Example: WINE (Windows applications on Linux).
- **Application Level** – Separates applications from the underlying OS, allowing them to run in isolated environments, improving security and compatibility. Examples: Citrix Virtual Apps, Microsoft App-V.

Load Balancing and Virtualization

- One characteristic of cloud computing is **virtualized network access to a service.** No matter where you access the service, you are directed to the available resources. The technology used to distribute service requests to resources is referred to as **load balancing.**

- Load balancing can be implemented in hardware, as is the case with F5's BigIP servers, or in software, such as the Apache mod_proxy_balancer extension, the Pound load balancer and reverse proxy software, and the Squid proxy and cache daemon.

- Load balancing is an optimization technique; it can be used to increase utilization and throughput, lower latency, reduce response time, and avoid system overload.

The following network resources can be load balanced:

- Network interfaces and services such as DNS, FTP, and HTTP
- Connections through intelligent switches
- Processing through computer system assignment
- Storage resources
- Access to application instances

Load Balancing and Virtualization

- Without load balancing, cloud computing would very difficult to manage. Load balancing provides the necessary redundancy to make an intrinsically unreliable system reliable through managed redi-rection.
- It also provides fault tolerance when coupled with a failover mechanism. Load balancing is nearly always a feature of server farms and computer clusters and for high availability applications.
- A load-balancing system can use different mechanisms to assign service direction. In the simplest load-balancing mechanisms, the load balancer listens to a network port for service requests.
- When a request from a client or service requester arrives, the load balancer uses a scheduling algorithm to assign where the request is sent.
- Scheduling algorithms in use today are **round robin and weighted round robin,** fastest response time, least connections and weighted least connections, and custom assignments based on other factors.

Load Balancing and Virtualization

- A session ticket is created by the load balancer so that subsequent related traffic from the client that is part of that session can be properly routed to the same resource.
- Without this session record or persistence, a load balancer would not be able to correctly failover a request from one resource to another.
- Persistence can be enforced using session data stored in a database and replicated across multiple load balancers. Other methods can use the client's browser to store a client-side cookie or through the use of a rewrite engine that modifies the URL.
- A session cookie stored on the client has the least amount of overhead for a load balancer because it allows the load balancer an independent selection of resources.
- The algorithm can be based on a simple round robin system where the next system in a list of systems gets the request.
- Round robin DNS is a common application, where IP addresses are assigned out of a pool of available IP addresses. Google uses round robin DNS,

Load Balancing and Virtualization

Advanced load balancing :

- The more sophisticated load balancers are workload managers. They determine the current utilization of the resources in their pool, the response time, the work queue length, connection latency and capacity, and other factors in order to assign tasks to each resource.

- Among the features you find in load balancers are polling resources for their health, the ability to bring standby servers online (priority activation), workload weighting based on a resource's capacity HTTP traffic compression, TCP offload and buffering, security and authentication, and packet shaping using content filtering and priority queuing.

- An Application Delivery Controller (ADC) is a combination load balancer and application server that is a server placed between a firewall or router and a server farm providing Web services.

- An Application Delivery Controller is assigned a virtual IP address (VIP) that it maps to a pool of servers based on application specific criteria.

- An ADC is a combination network and application layer device. You also may come across ADCs referred to as a content switch, multilayer switch, or Web switch.

Load Balancing and Virtualization

Vendors, among others, sell ADC systems :

- A10 Networks (http://www.a10networks.com/)
- Barracuda Networks (http://www.barracudanetworks.com/)
- Brocade Communication Systems (http://www.brocade.com/)
- Cisco Systems (http://www.cisco.com/)
- Citrix Systems (http://www.citrix.com/)
- 5 Networks (http://www.f5.com/)
- Nortel Networks (http://www.nortel.com/)
- Coyote Point Systems (http://www.coyotepoint.com/)
- Radware (http://www.radware.com/)

Load Balancing and Virtualization

- Services provided by an ADC include data compression, content caching, server health monitoring, security, SSL offload and advanced routing based on current conditions.
- An ADC is considered to be an application accelerator, and the current products in this area are usually focused on two areas of technology: network optimization, and an application or frame work optimization.
- An architectural layer containing ADCs is described as an **Application Delivery Network (ADN),** and is considered to provide WAN optimization services. Often an ADN is comprised of a pair of redundant ADCs.
- The purpose of an ADN is to distribute content to resources based on application specific criteria. ADN provide a caching mechanism to reduce traffic, traffic prioritization and optimization, and other techniques.
- ADN began to be deployed on **Content Delivery Networks (CDN)** in the late 1990s, where it added the ability to optimize applications (application fluency) to those networks.
- Most of the ADC vendors offer commercial ADN solutions.

Load Balancing and Virtualization

Feature	ADN	CDN
Focus	Applications	Static content
Traffic type	Dynamic + static	Mostly static
Location	Data center / cloud	Edge locations
Functions	LB, security, optimization	Caching, faster delivery

Load Balancing and Virtualization

Additional ADN vendors :

- Akamai Technologies (http://www.akamai.com/)
- Blue Coat Systems (http://www.bluecoat.com/)
- CDNetworks (http://www.cdnetworks.com/)
- Crescendo Networks (http://www.crescendonetworks.com/)
- Expand Networks (http://www.expand.com/)
- Juniper Networks (http://www.juniper.net/)
- Google's cloud is a good example of the use of load balancing.

Load Balancing and Virtualization

The Google cloud :

- According to the Web site tracking firm Alexa (http://www.alexa.com/topsites), Google is the single most heavily visited site on the Internet; that is, Google gets the most hits.
- The investment Google has made in infrastructure is enormous, and the Google cloud is one of the largest in use today. It is estimated that Google runs over a million servers worldwide, processes a billion search requests, and generates twenty petabytes of data per day.
- Google is understandably reticent to disclose much about its network, because it believes that its infrastructure, system response, and low latency are key to the company's success.
- Google never gives datacenter tours to journalists, doesn't disclose where its datacenters are located, and obfuscates the locations of its datacenters by wrapping them in a corporate veil.
- The discretely named Tetra **LLC (limited liability company)** owns the land for the Council Bluffs, Iowa, site, and Lapis LLC owns the land for the Lenoir, North Carolina, site.

Load Balancing and Virtualization

- This makes Google infrastructure watching something akin to a sport to many people.
- Google's infrastructure and the basic idea behind how Google distributes its traffic by pooling IP addresses and performing several layers of load balancing.
- Google has many datacenters around the world. As of March 2008, Rich Miller of DataCenterKnowledge.com wrote that Google had at least 12 major installations in the United States and many more around the world.
- Google supports over 30 country specific versions of the Google index, and each localization is supported by one or more datacenters. For example, Paris, London, Moscow, Sao Paolo, Tokyo, Toronto, Hong Kong, Beijing and others support their countries' locale.
- Germany has three centers in Berlin, Frankfurt, and Munich; the Netherlands has two at Groningen and Eemshaven.
- The countries with multiple datacenters store index replicas and support network peering relationships. Network peering helps Google have low latency connections to large Internet hubs run by different network providers.

Load Balancing and Virtualization

- Google's datacenters are sited based on the following factors (roughly in order of importance):
- Availability of cheap and, if possible, renewable energy
- The relative locations of other Google datacenters such that the site provides the lowest latency response between sites
- Location of nearby Internet hubs and peering sites
- A source of cooling water
- The ability to purchase a large area of land surrounding the site
- Speculation on why Google purchases large parcels of land ranges from creating a buffer zone between the datacenter and surrounding roads and towns or possibly to allow for building wind farms when practical.
- Tax concessions from municipalities that lower Google's overhead.

Load Balancing and Virtualization

- **Google maintains a pool of hundreds of IP addresses,** all of which eventually resolve to its Mountain View, California, headquarters.
- When you initiate a Google search, your query is sent to a DNS server, which then queries Google's DNS servers.
- The Google DNS servers examine the pool of addresses to determine which addresses are geographically closest to the query origin and uses a round robin policy to assign an IP address to that request.
- The request usually goes to the nearest datacenter, and that IP address is for a cluster of Google servers. This DNS assignment acts as a first level of IP virtualization, a pool of network addresses have been load balanced based on geography.
- **A Google cluster can contain thousands of server**s. Google servers are racks of commodity 1U or 2U servers containing 40 to 80 servers per rack with one switch per rack.
- Each switch is connected to a core gigabit switch. Google servers run a customized version of Linux with applications of several types.

Load Balancing and Virtualization

- When the query request arrives at its destination, a Google cluster is sent to a load balancer, which forwards that request to a Squid proxy server and Web cache dameon. This is the second level of IP distribution, based on a measure of the current system loading on proxy servers in the cluster.
- The Squid server checks its cache, and if it finds a match to the query, that match is returned and the query has been satisfied.
- If there is no match in the Squid cache, the query is sent to an individual Google Web Server based on current Web server utilizations, which is the third level of network load balancing, again based on utilization rates.
- It is the Google Web Servers that perform the query against the Google index and then format the results into an HTML page that is returned to the requester.
- This procedure then performs two more levels of load balancing based on utilization rates. Google's secret sauce is its in-memory inverted index and page rank algorithm.
- Google's GoogleBot crawls the Web and collects document information. Some details of the search and store algorithm are known. Google looks at the title and first few hundred words and builds a word index from the result. Indexes are stored on an index server.

Load Balancing and Virtualization

- Some documents are stored as snapshots (PDF, DOC, XLS, and so on), but lots of information is not addressed in the index.
- Each document is given a unique ID ("docid"), and the content of the document is disassembled into segments called shards, subjected to a data compression scheme and stored on a document server.
- The entire index is maintained in system memory partitioned over each instance of the index's replicas. A page rank is created based on the significant links to that page.
- Queries are divided into word lists, and the Google algorithm examines the words and the relationships of one word to another.
- Those word relationships are mapped against the main index to create a list of documents, a feature called an inverted index. In an inverted index, words are mapped to documents, which can be done very quickly when the index is fully kept in memory.
- The Web server takes the result of a query and composes the Web page from that result. Ads included on the page are from ad servers, which provide Google's AdSense and AdWords services.

Load Balancing and Virtualization

- The query also is presented to a spelling server to provide suggestions for alternative spellings to include in the search result.
- Certain keywords, data input patterns, and other strings are recognized as having special operational significance.
- For example entering “2 plus 2” initiates Google’s calculator program, and a ten-digit number returns a reverse phone lookup using the phonebook program. These programs are supported by special application servers.
- Google doesn’t use hardware virtualization; it performs server load balancing to distribute the processing load and to get high utilization rates.
- The workload management software transfers the workload from a failed server over to a redundant server, and the failed server is taken offline.
- Multiple instances of various Google applications are running on different hosts, and data is stored on redundant storage systems.

Load Balancing and Virtualization

Aspect	Virtualization	Load Balancing
Primary function	Creates virtual servers/resources	Distributes traffic/workload
Layer of operation	Hardware / OS level	Network / Transport / Application level
Main component	Hypervisor	Load balancer (hardware/software)
User visibility	Mostly invisible to end users	Directly affects user experience
Failure handling	VM isolation & migration	Traffic redirection & failover
Scalability type	Vertical & horizontal (add VMs)	Horizontal (add/remove servers)
Security role	Basic isolation	Advanced (WAF, DDoS, SSL)
Performance impact	Slight overhead	Improves performance
Deployment dependency	Needs host hardware	Needs backend servers
Management focus	Resource management	Traffic management

Load Balancing and Virtualization

Category	Example Players
Enterprise Hypervisors (server virtualization)	VMware, Microsoft, Oracle, IBM, Red Hat
Cloud Virtualization / IaaS	AWS, Google Cloud, Microsoft Azure
Desktop & App Virtualization	Citrix, VMware Horizon, Microsoft (Azure Virtual Desktop)
Hyper-Converged / Infrastructure Platforms	Nutanix, HPE, Cisco, Dell

Understanding Hypervisors

- Load balancing virtualizes systems and resources by mapping a logical address to a physical address. Another fundamental technology for abstraction creates virtual systems out of physical systems.
- If load balancing is like playing a game of hot potato, then virtual machine technologies is akin to playing slice and dice with the potato.
- Given a computer system with a certain set of resources, you can set aside portions of those resources to create a virtual machine.
- The standpoint of applications or users, a virtual machine has all the attributes and characteristics of a physical system but is strictly software that emulates a physical machine.
- A system virtual machine (or a hardware virtual machine) has its own address space in memory, its own processor resource allocation, and its own device I/O using its own virtual device drivers.
- Virtual machines are designed to run only a single application or process and are referred to as process virtual machines.

Understanding Hypervisors

- A virtual machine is a computer that is walled off from the physical computer that the virtual machine is running on
- This makes virtual machine technology very useful for running old versions of operating systems.
- Testing applications in what amounts to a sandbox, or in the case of cloud computing, creating virtual machine instances that can be assigned a workload.
- Virtual machines provide the capability of running multiple machine instances, each with their own operating system.
- VMMs to manage application provisioning, provide for machine instance cloning and replication, allow for graceful system failover, and provide several other desirable features.
- The downside of virtual machine technologies is that having resources indirectly addressed means there is some level of overhead.

Understanding Hypervisors

Virtual machine types :

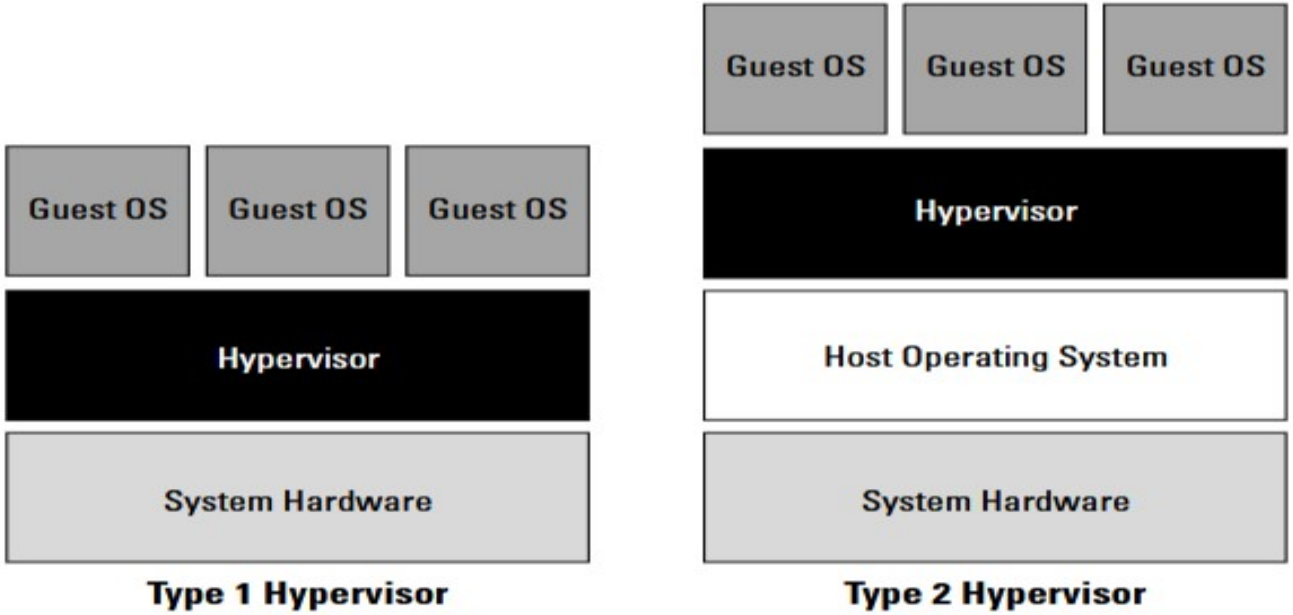
- A low-level program is required to provide system resource access to virtual machines, and this program is referred to as the hypervisor or Virtual Machine Monitor (VMM).
- A hypervisor running on bare metal is a Type 1 VM or native VM.
- **Examples of Type 1 Virtual Machine Monitors are LynxSecure, RTS Hypervisor, Oracle VM, Sun xVM Server, VirtualLogix VLX, VMware ESX and ESXi, and Wind River VxWorks, among others**.
- The operating system loaded into a virtual machine is referred to as the guest operating system, and there is no constraint on running the same guest on multiple VMs on a physical system.
- Type 1 VMs have no host operating system because they are installed on a bare system.
- An operating system running on a Type 1 VM is a full virtualization because it is a complete simulation of the hardware that it is running on.

Understanding Hypervisors

- Some hypervisors are installed over an operating system and are referred to as Type 2 or hosted VM.
- **Examples of Type 2 Virtual Machine Monitors are Containers, KVM, Microsoft Hyper V, Parallels Desktop for Mac, Wind River Simics, VMWare Fusion, Virtual Server 2005 R2, Xen, Windows Virtual PC, and VMware Workstation 6.0 and Server, among others.**
- This is a very rich product category. Type 2 virtual machines are installed over a host operating system; for Microsoft Hyper-V, that operating system would be Windows Server.
- In the section that follows, the Xen hypervisor (which runs on top of a Linux host OS) is more fully described. Xen is used by AmazoN Web Services to provide Amazon Machine Instances (AMIs). On a Type 2 VM, a software interface is created that emulates the devices with which a system would normally interact.
- This abstraction is meant to place many I/O operations outside the virtual environment, which makes it both programmatically easier and more efficient to execute device.
- I/O than it would be inside a virtual environment. This type of virtualization is sometimes referred to as para virtualization, and it is found in hypervisors such as Microsoft's Hyper-V and Xen. It is the host operating system that is performing the I/O through a para-API.

Understanding Hypervisors

VMware's vSphere cloud computing infrastructure model



Understanding Hypervisors

Feature	Type 1 Hypervisor	Type 2 Hypervisor
Also called	Bare-metal / Native	Hosted
Runs on	Directly on hardware	On top of host OS
Boot process	Loads at system startup	Starts after host OS
Host OS required	No	Yes
Performance	Very high	Moderate to low
Security	More secure	Less secure
Resource overhead	Low	High
Hardware access	Direct	Through host OS
Latency	Low	Higher
Scalability	High	Limited
Stability	Very stable	Depends on host OS
Failure impact	Limited to VM	Host OS failure affects all VMs
Use case	Data centers, cloud, servers	Desktop, testing, learning
Installation	Complex	Simple
Cost	Usually enterprise-level	Often free or low cost
Examples	ESXi, Hyper-V, Xen, KVM	VirtualBox, VMware Workstation

Understanding Hypervisors

- The difference between emulation, para virtualization, and full virtualization. In emulation, the virtual machine simulates hardware, so it can be independent of the underlying system hardware. A guest operating system using emulation does not need to be modified in any way.

- Para virtualization requires that the host operating system provide a virtual machine interface for the guest operating system and that the guest access hardware through that host VM.

- An operating system running as a guest on a para virtualization system must be ported to work with the host interface. Finally, in a full virtualization scheme, the VM is installed as a Type 1 Hypervisor directly onto the hardware.

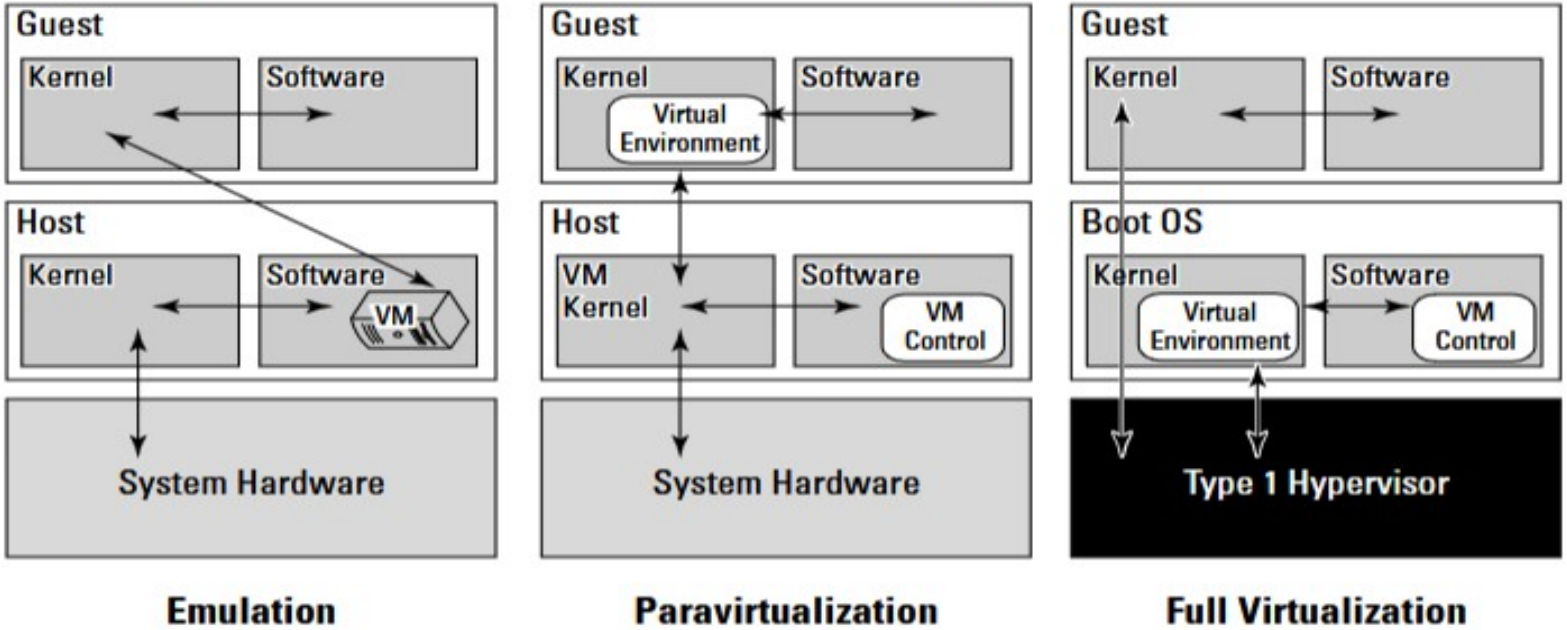
- All operating systems in full virtualization communicate directly with the VM hypervisor, so guest operating systems do not require any modification.

- Guest operating systems in full virtualization systems are generally faster than other virtualization schemes.

- The Virtual Machine Interface (VMI) open standard (http://vmi.ncsa.uiuc.edu/) that VMware has proposed is an example of a para virtualization API.

Understanding Hypervisors

Emulation, paravirtualization, and full virtualization types



Emulation **Paravirtualization** **Full Virtualization**

Understanding Hypervisors

- **Emulation** is a technique in which **one system imitates the hardware and software of another system**, allowing programs designed for one architecture to run on a **completely different architecture**.

Key Points :

- Emulates **entire hardware environment**
- Guest system believes it is running on real hardware
- **No dependency** on host CPU architecture
- Very **slow** compared to virtualization
- Does **not require special hardware support**
- Useful for running **legacy or foreign OS/software**
- High flexibility, low performance

Understanding Hypervisors

Feature	Emulation	Virtualization
Hardware imitation	Complete	Partial or none
Guest OS modification	Not required	Not required (full virt)
CPU architecture	Can be different	Must be same
Performance	Very slow	Fast
Hardware support	Not needed	Often required
Use case	Legacy / foreign systems	Servers, cloud, testing

Understanding Hypervisors

- **Paravirtualization:** Guest OS is modified to interact directly with the hypervisor.
- **Full Virtualization:** Guest OS runs unmodified on fully emulated hardware.

Feature	Paravirtualization	Full Virtualization
Guest OS modification	Required	Not required
OS awareness	Aware of hypervisor	Not aware
Performance	Higher	Lower
Hardware emulation	Minimal	Complete
Compatibility	Limited	High
Overhead	Low	High
Security	High	High
Use case	High-performance servers	General-purpose virtualization
Examples	Xen (PV mode)	VMware, VirtualBox

Understanding Hypervisors

- Applications running inside an application virtual machine are generally slow, but these programs are very popular because they provide portability, offer rich programming languages, come with many advanced features, and allow platform independence for their programs.
- Cloud computing applications provide process virtual machine applications, this type of abstraction isn't really suitable for building a large or high-performing cloud network, with one exception.
- The exception is the process VMs that enable a class of parallel cluster computing applications.
- These applications are high-performance systems where the virtual machine is operating one process per cluster node, and the system maintains the necessary intra-application communications over the network interconnect.
- Examples of this type of system are the Parallel Virtual Machine (PVM; see http://www.csm.ornl.gov/pvm/pvm_home.html)
- The Message Passing Interface (MPI; see http://www.mpi-forum.org/). Some people do not consider these application VMs to be true virtual machines, noting that these applications can still access the host operating system services on the specific system on which they are running.

Understanding Hypervisors

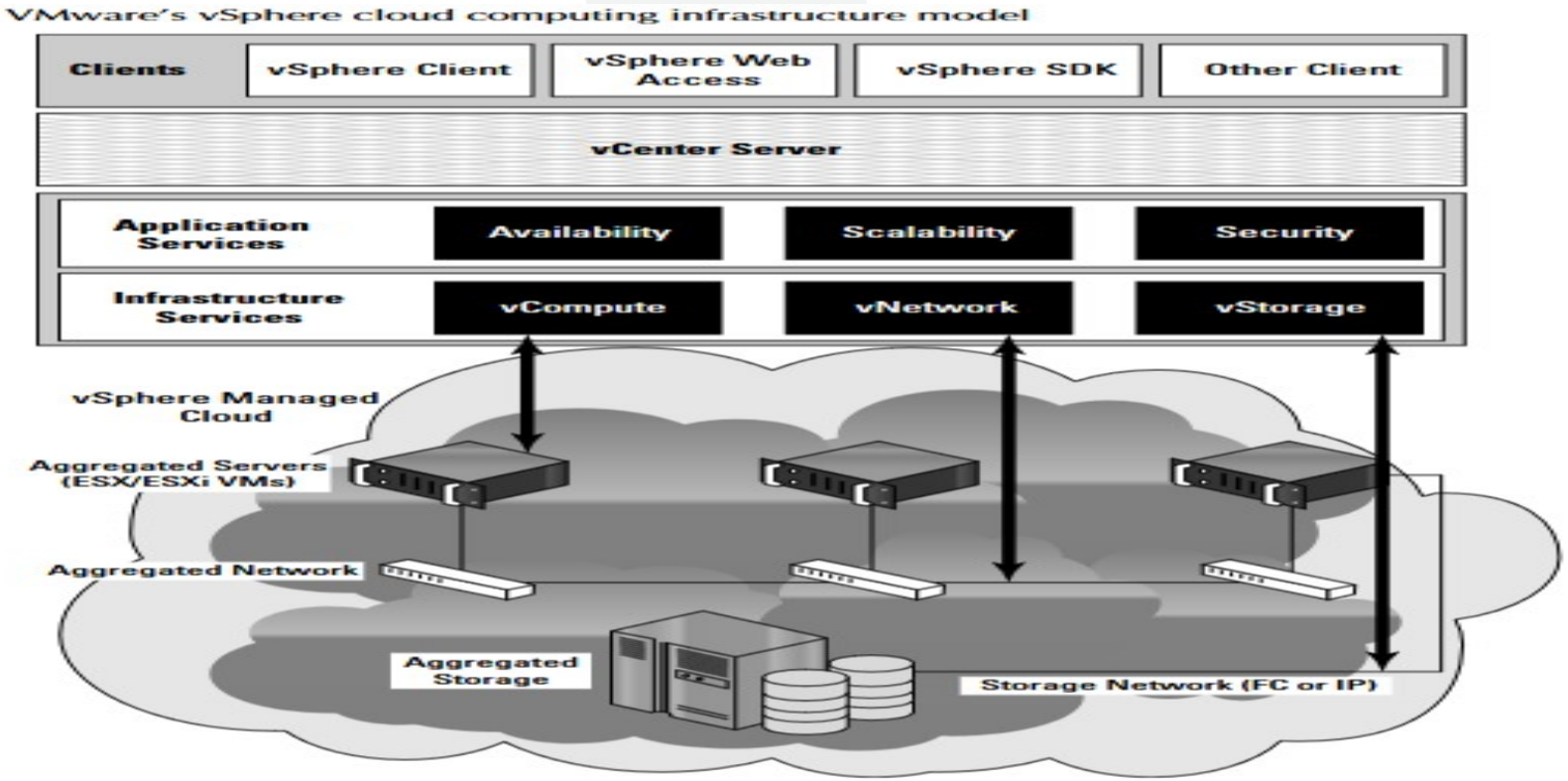
- Some operating systems such as Sun Solaris and IBM AIX 6.1 support a feature known as operating system virtualization.
- This type of virtualization creates virtual servers at the operating system or kernel level. Each virtual server is running in its own virtual environment (VE) as a virtual private server (VPS).
- Different operating systems use different names to describe these machine instances, each of which can support its own guest OS.
- However, unlike true virtual machines, VPS must all be running the same OS and the same version of that OS.
- Sun Solaris 10 uses VPS to create what is called Solaris Zones. With IBM AIX, the VPS is called a System Workload Partition (WPAR).
- This type of virtualization allows for a dense collection of virtual machines with relatively low overhead.

Understanding Hypervisors

VMware vSphere :

- VMware vSphere is a management infrastructure framework that virtualizes system, storage, and networking hardware to create cloud computing infrastructures.
- vSphere is the branding for a set of management tools and a set of products previously labeled VMware Infrastructure.
- vSphere provides a set of services that applications can use to access cloud resources, including these:
- VMware vCompute: A service that aggregates servers into an assignable pool
- VMware vStorage: A service that aggregates storage resources into an assignable pool
- VMware vNetwork: A service that creates and manages virtual network interfaces
- Application services: Such as HA (High Availability) and Fault Tolerance
- vCenter Server: A provisioning, management, and monitoring console for VMware cloud infrastructures

Understanding Hypervisors



Understanding Hypervisors

- A vSphere cloud is a pure infrastructure play. The virtualization layer that abstracts processing, memory, and storage uses the VMware ESX or ESXi virtualization server.
- ESX is a Type 1 hypervisor; it installs over bare metal (a clean system) using a Linux kernel to boot and installs the vmkernel hypervisor (virtualization kernel and support files).
- When the system is rebooted, the vmkernel loads first, and then the Linux kernel becomes the first guest operating system to run as a virtual machine on the system and contains the service console.
- VMware is a very highly developed infrastructure and the current leader in this industry. A number of important add-on products are available for cloud computing applications.

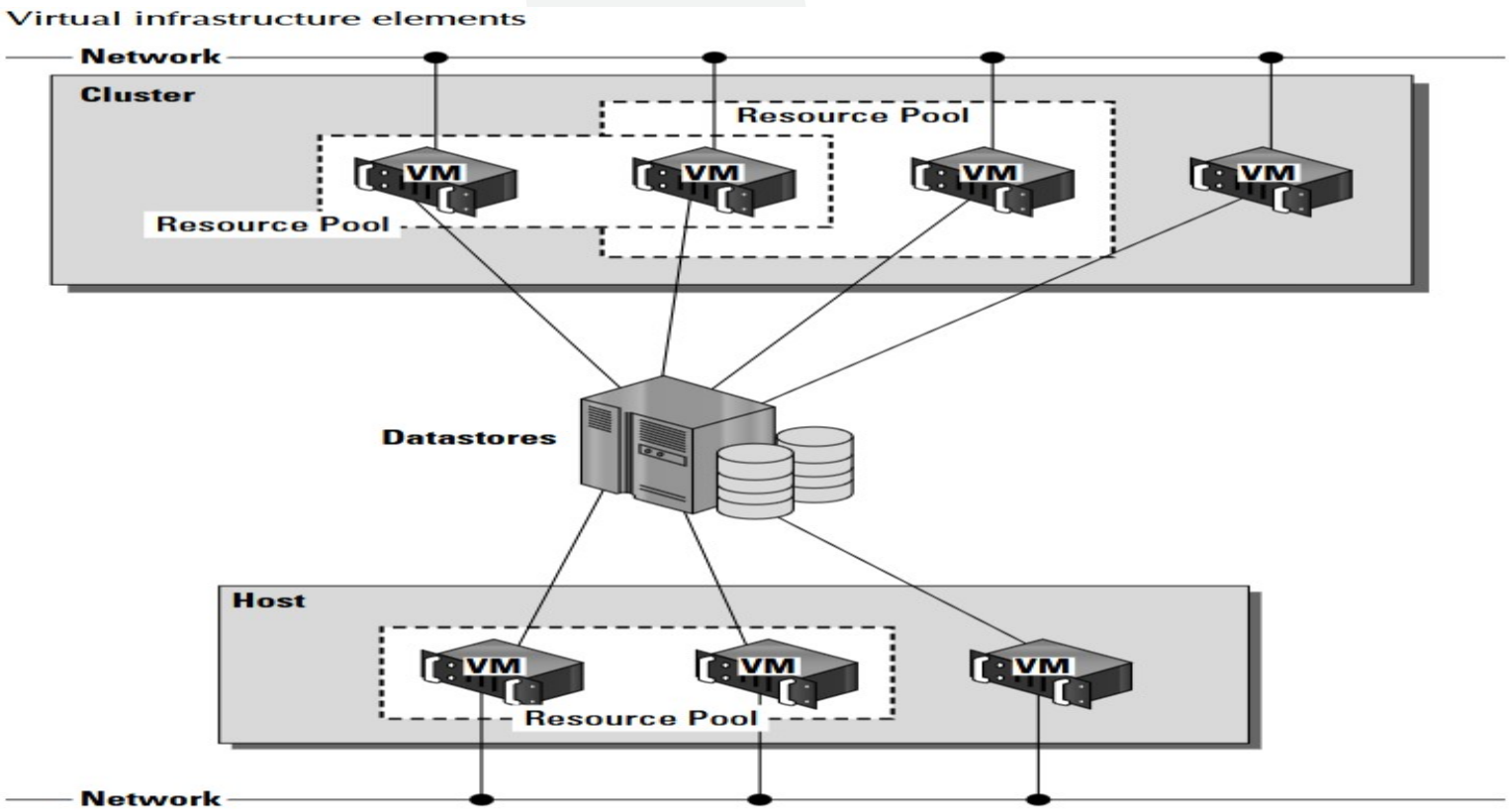
These are among the more notable products:

- **Virtual Machine File System (VMFS):** A high-performance cluster file system for an ESX/ESXi cluster.
- **VMotion:** A service that allows for the migration of a virtual machine from one physical server to another physical server while the virtual server runs continuously and without any interruption of ongoing transactions.

Understanding Hypervisors

- **Storage VMotion:** A product that can migrate files from one datastore to another data store while the virtual machine that uses the datastore continues to run.
- **Virtual SMP:** A feature that allows a virtual machine to run on two or more physical processors at the same time.
- **Distributed Resource Scheduler (DRS):** A system for provisioning virtual machines and load balancing processing resources dynamically across the different physical systems that are in use.
- A part of the DRS called the distributed power management (DPM) module can manage the power consumption of systems.
- **vNetwork Distributed Switch (DVS):** A capability to maintain a network runtime state for virtual machines as they are migrated from one physical system to another.
- **DVS also monitors network** connections, **provides firewall services, and enables the use of third party switches such as the Cisco Nexus 1000V to manage virtual networks**.

Understanding Hypervisors



Understanding Hypervisors

- Storage resources can be either **Direct Attached Storage (DAS)** of a server using SCSI, SAS, or Serial Advanced Technology Attachment (SATA) connections, Fibre Channel disk arrays/SANs, iSCSI disk arrays/SANs, or Network Attached Storage (NAS) disk arrays.

- Although the lines drawn between the data store and different VMs indicate a direct connection, with the exception of DAS, the other storage types are shared storage solutions.

- Storage virtualization is most commonly achieved through a mapping mechanism where a logical storage address is translated into a physical storage address. Block-based storage such as those used.

- In SANs use a feature called a **Logical Unit Identifier (LUN) with specific addresses stored in the form of an offset called the Logical Block Address (LBA).** The address space mapping then maps the address of the logical or virtual disk (vdisk) to the logical unit on a storage controller.

- Storage virtualization may be done in software or in hardware, and it allows requests for virtualized storage to be redirected as needed.
- Network virtualization abstracts networking hardware and software into a virtual network that can be managed. A virtual network can create virtual network interfaces **(VNICs) or virtual LANs (VLANS)** and can be managed by a hypervisor, operating system, or external management console.

Understanding Hypervisors

Feature	SATA	SAS
Target market	Consumer / low-cost servers	Enterprise servers
Speed	Up to 6 Gbps	Up to 12 Gbps (SAS-3)
Reliability	Single-port	Dual-port, better error correction
Hot swap	Optional	Standard
Cost	Low	Higher

Term	What it is	Example
SCSI	Command protocol	Read / Write commands
SAS	Transport/interface	Cable + controller
iSCSI	SCSI over IP	Ethernet SAN
FC	SCSI over Fibre Channel	Fibre SAN

Understanding Machine Imaging

- The abstractions that cloud computing needs can be **achieved through redirection and virtualization.** A third mechanism is commonly used to provide system portability, instantiate applications, and provision and deploy systems in the cloud.

- This third mechanism is through storing the state of a systems using a system image. A system image makes a copy or a clone of the entire computer system inside a single container such as a file.

- The system imaging program is used to make this image and can be used later to restore a system image. Some imaging programs can take snapshots of systems, and most allow you to view the files contained in the image and do partial restores.

- A prominent example of a system image and how it can be used in cloud computing architectures is the Amazon Machine Image (AMI) used by Amazon Web Services to store copies of a virtual machine.

- When you subscribe to AWS, you can choose to use one of its hundreds of canned AMIs or to create a custom system and capture that system's image to an AMI.
- An **AMI can be for public use under a free distribution license, for pay-per-use with operating systems** such as Windows, or shared by an EC2 user with other users who are given the privilege of access.

Porting Applications

- Cloud computing applications have the ability to run on virtual systems and for these systems to be moved as needed to respond to demand.
- Systems (VMs running applications), storage, and network assets can all be virtualized and have sufficient flexibility to give acceptable distributed WAN application performance.
- Developers who write software to run in the cloud will undoubtedly want the ability to port their applications from one cloud vendor to another, but that is a much more difficult proposition.
- Cloud computing is a relatively new area of technology, and the major vendors have technologies that don't interoperate with one another.

The Simple Cloud API :

- If you build an application on a platform such as Microsoft Azure, porting that application to Amazon Web Services or GoogleApps may be difficult, if not impossible.
- In an effort to create an interoperability standard, Zend Technologies has started an open source initiative to create a common application program interface that will allow applications to be portable.

Porting Applications

- The initiative is called the **Simple API for Cloud Application Services (http://www.simplecloud.org/),** and the effort has drawn interest from several major cloud computing companies.

- Among the founding supporters are IBM, Microsoft, Nivanix, Rackspace, and GoGrid. Simple Cloud API has as its goal a set of common interfaces for:

- **File Storage Services:** Currently Amazon S3, Windows Azure Blob Storage, Nirvanix and Local storage is supported by the Storage API.

- **Document Storage Services**: Amazon SimpleDB and Windows Azure Table Storage are currently supported. Local document storage is planned.

- **Simple Queue Services**: Amazon SQS, Windows Azure Queue Storage, and Local queue services are supported. Zend intends to add the interface to their open source PHP Framework (http://www.frame work.zend.com) as the Zend_Cloud framework component.

- Vendors such as Microsoft and IBM are supplying adapters that will use part of the Simple Cloud API for their cloud application services.

AppZero Virtual Application Appliance

- Applications that run in datacenters are captive to the operating systems and hardware platforms that they run on. Many datacenters are a veritable Noah's Ark of computing.
- Moving an application from one platform to another isn't nearly as simple as moving a machine image from one system to another.
- The situation is further complicated by the fact that applications are tightly coupled with the operating systems on which they run. An application running on Windows, for example, isn't isolated from other applications.
- When the application loads, it often loads or uses different Dynamic Link Libraries (DLL), and it is through the sharing or modification of DLLs that Windows applications get themselves in trouble.
- Further modifications include modifying the registry during installation. These factors make it difficult to port applications from one platform to another without lots of careful work.
- If you are a Platform as a Service (PaaS) application developer, you are packaging a complete software stack that includes not only your application, but the operating system and application logic and rules as well. Vendor lock-in for you application is assured.

AppZero Virtual Application Appliance

- The ability to run an application from whatever platform you want is not one of the characteristics of cloud computing, but you can imagine that it is a very attractive proposition.
- While the Simple Cloud API is useful for applications written in PHP, other methods may be needed to make applications easily portable.
- One company working on this problem is AppZero (http://www. appzero.com/), and its solution is called the Virtual Application Appliance (VAA).
- The AppZero solution creates a virtual application appliance as an architectural layer between the Windows or the UNIX operating system and applications.
- The virtualization layer serves as the mediator for file I/O, memory I/O, and application calls and response to DLLs, which has the effect of sandboxing the application.
- The running application in AppZero changes none of the registry entries or any of the files on the Windows Server.

AppZero Virtual Application Appliance

- VAA creates a container that encapsulates the application and all the application's dependencies within a set of files; it is essentially an Application Image for a specific OS.
- Dependencies include DLL, service settings, necessary configuration files, registry entries, and machine and network settings.
- This container forms an installable server-side application stack that can be run after installation, but has no impact on the underlying operating system.
- VAAs are created using the AppZero Creator wizard, managed with the AppZero Admin tool, and may be installed using the AppZero Director, which creates a VAA runtime application.
- If desired, an application called AppZero Dissolve removes the VAA virtualization layer from the encapsulated application and installs that application directly into the operating system.
- AppZero envisages using VAAs to create what it calls a stateless cloud. In a stateless cloud, the application's state information is stored on a network share where it is available to run on different cloud systems as needed.

Capacity Planning

- Capacity planning for cloud computing sounds like an oxymoron. Why bother doing capacity planning for a resource that is both ubiquitous and limitless? **Capacity planning = "Right resources, right time, right place"**

- The reality of cloud computing is rather different than the ideal might suggest; cloud computing is neither ubiquitous, nor is it limitless.

- Capacity planning for a cloud computing system offers you many enhanced capabilities and some new challenges over a purely physical system.

- A capacity planner seeks to meet the future demands on a system by providing the additional capacity to fulfill those demands. Many people equate capacity planning with system optimization but they are not the same.

- System optimization aims to get more production from the system components you have. Capacity planning measures the maximum amount of work that can be done using the current technology and then adds resources to do more work as needed.

- If system optimization occurs during capacity planning, that is all to the good; but capacity planning efforts focus on meeting demand. If that means the capacity planner must accept the inherent inefficiencies in any system.

Capacity Planning

Capacity planning is an iterative process with the following steps:

1. Determine the characteristics of the present system.

2. Measure the workload for the different resources in the system: CPU, RAM, disk, network, and so forth.

3. Load the system until it is overloaded, determine when it breaks, and specify what is required to maintain acceptable performance.

Knowing when systems fail under load and what factor(s) is responsible for the failure is the critical step in capacity planning.

4. Predict the future based on historical trends and other factors.

5. Deploy or tear down resources to meet your predictions.

6. Iterate Steps 1 through 5 repeatedly.

Defining Baseline and Metrics

- The first item of business is to determine the current system capacity or workload as a measurable quantity over time.
- Many developers create cloud-based applications and Web sites based on a LAMP solution stack.

LAMP stands for :

- Linux, the operating system
- Apache HTTP Server (http://httpd.apache.org/), the Web server based on the work of the Apache Software Foundation
- MySQL (http://www.mysql.com), the database server developed by the Swedish company MySQL AB, owned by Oracle Corporation through its acquisition of Sun Microsystems.
- PHP (http://www.php.net/), the Hypertext Preprocessor scripting language developed by The PHP Group.

Defining Baseline and Metrics

- Four technologies are open source products, although the distributions used may vary from cloud to cloud and from machine instance to machine instance. A **baseline** is a **record of normal or typical performance** for a system, based on collected metrics over time.
- On Amazon Web Services, machine instances are offered for both **Red Hat Linux and for Ubuntu.**
- LAMP stacks based on Red Hat Linux are more common than the other distributions, but SUSE Linux and Debian GNU/Linux are also common.
- Variants of LAMP are available that use other operating systems such as the **Macintosh, OpenBSD (OpAMP), Solaris (SAMP), and Windows (WAMP).**
- Although many other common applications are in use, LAMP is good to use as an example because it offers a system with two applications **(APACHE and MySQL)** that can be combined or run separately on servers.
- LAMP is one of the major categories of Amazon Machine Instances that you can create on the Amazon Web Service.

Baseline Measurements

- A capacity planner is working with a system that has a Web site based on APACHE, and let's assume the site is processing database transactions using MySQL. **Baseline measurements** are the **actual recorded values of system performance and resource usage** over a specific period. They are used to define what is **"normal"** for your environment.

Two important overall workload metrics in this LAMP system:

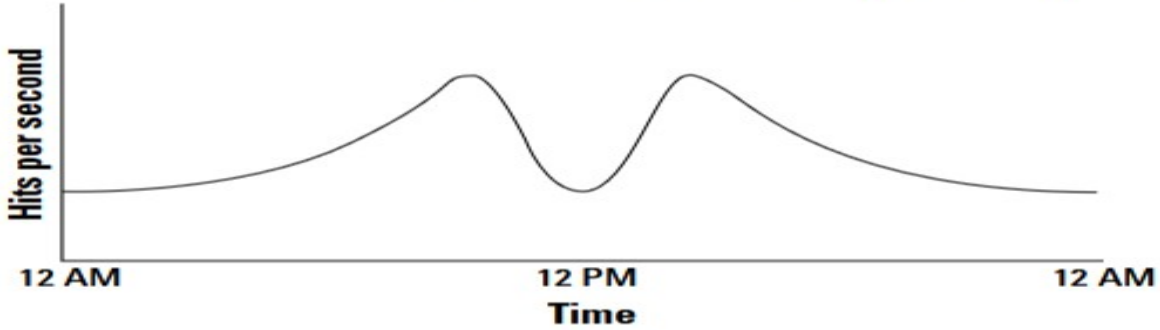
- **Page views** or hits on the Web site, as measured in hits per second.
- **Transactions** completed on the database server, as measured by transactions per second or perhaps by queries per second.
- The historical record for the Web server page views over a hypothetical day, week, and year are graphed.
- These graphs are created by summing the data from the different servers contained in the performance Web logs of the site or measuring throughput from the system based on an intervening service.
- A typical daily log (shown on top) shows two spikes in demand, one that is centered around 10 AM EST when users on the east coast of the United States use the site heavily and another at around 1 PM EST.
- When users on the west coast of the United States use the site heavily. The three-hour time difference is also the difference between Eastern Standard Time and Pacific Standard Time.

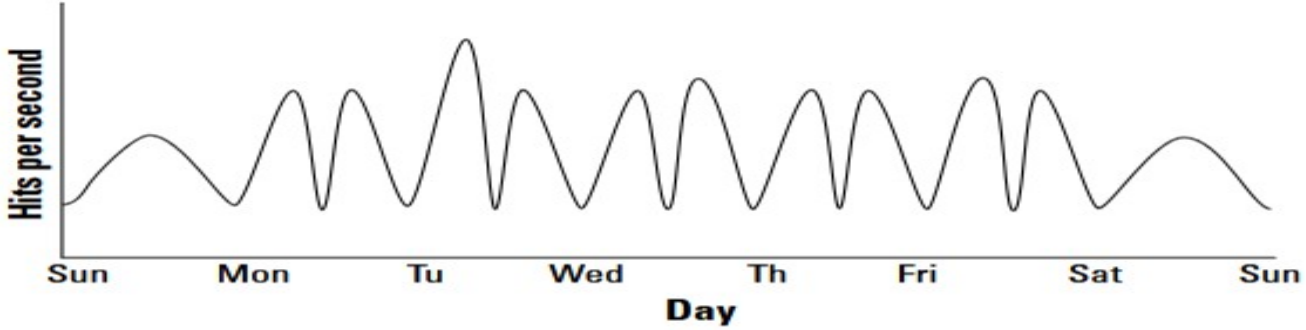
Baseline Measurements

- These daily performance spikes occur on workdays, but the spikes aren't always of equal demand on weekends. The demand rises through the middle of the day and then ebbs later.
- Some days show a performance spike, as shown on the Tuesday morning section of the weekly graph. A goal of capacity planning is to correlate these performance spikes and dips with particular events and causations.
- Also, at some point your traffic patterns may change, so you definitely want to evaluate these statistics on an ongoing basis.
- The yearly graph, at the bottom of Figure 6.1, shows that the daily averages and the daily peaks rise steadily over the year and, in fact, double from January 1st at the start of the year to December 31st at year end.
- Knowing this, a capacity planner would need to plan to serve twice the traffic while balancing the demands of peak versus average loads. However, it may not mean that twice the resources need to be deployed.
- The amount of resources to be deployed depends upon the characterization of the Web servers involved, their potential utilization rates, and other factors.

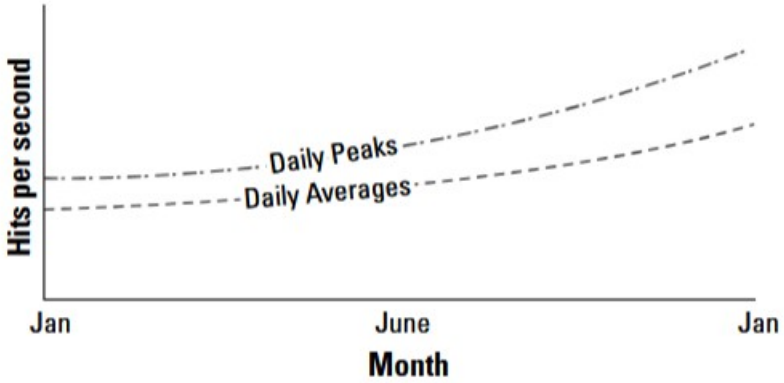
Baseline Measurements

A Web servers' workload measured on a day, a week, and over the course of a year





Baseline Measurements



A number of important characteristics are determined by these baseline studies:

- WT the total workload for the system per unit time To obtain WT you need to integrate the area under the curve for the time period of interest.
- WAVG, the average workload over multiple units of time To obtain WAVG you need to sum various WTs and divide by the number of unit times involved. You may also want to draw a curve that represents the mean work done.

Baseline Measurements

- WMAX the highest amount of work recorded by the system. This is the highest recorded system utilization.
- WTOT the total amount of work done by the system, which is determined by the sum of WT (ΣWT).
- A similar set of graphs would be collected to characterize the database servers, with the workload for those servers measured in transactions per second.
- As part of the capacity planning exercise, the workload for the Web servers would be correlated with the workload of the database servers to determine patterns of usage.
- The goal of a capacity planning exercise is to accommodate spikes in demand as well as the overall growth of demand over time.
- The growth in demand over time is the most important consideration because it represents the ability of a business to grow.
- A spike in demand may or may not be important enough to an activity to attempt to capture the full demand that the spike represents.

System Metrics

- Determined what amounts to application-level statistics for your site. These measurements are an aggregate of one or more Web servers in your infrastructure.
- Capacity planning must measure system-level statistics, determining what each system is capable of, and how resources of a system affect system-level performance.
- In some instances, the capacity planner may have some influence on system architecture, and the impact of system architecture on application- and system-level statistics may be examined and altered.

A machine instance is primarily defined by four essential resources:

- CPU
- Memory (RAM)
- Disk
- Network connectivity

System Metrics

1. CPU Metrics :

- CPU utilization (%) – How much CPU is being used
- Load average – Number of processes waiting for CPU
- Context switches – Frequency of task switching

2. Memory Metrics :

- Used / Free memory
- Cache and buffer usage
- Page faults / swap usage

3. Disk / Storage Metrics :

- Disk I/O (read/write rate)
- Disk latency
- Disk space usage (%)

System Metrics

4. Network Metrics

-
- Bandwidth usage
- Latency
- Packet loss
- Error rates

5. Application Metrics

- Response time
- Throughput (requests per second)
- Error rate
- Active sessions/users

6. Database Metrics :

- Query execution time
- Connections in use
- Lock waits / deadlocks
- Cache hit ratio

System Metrics

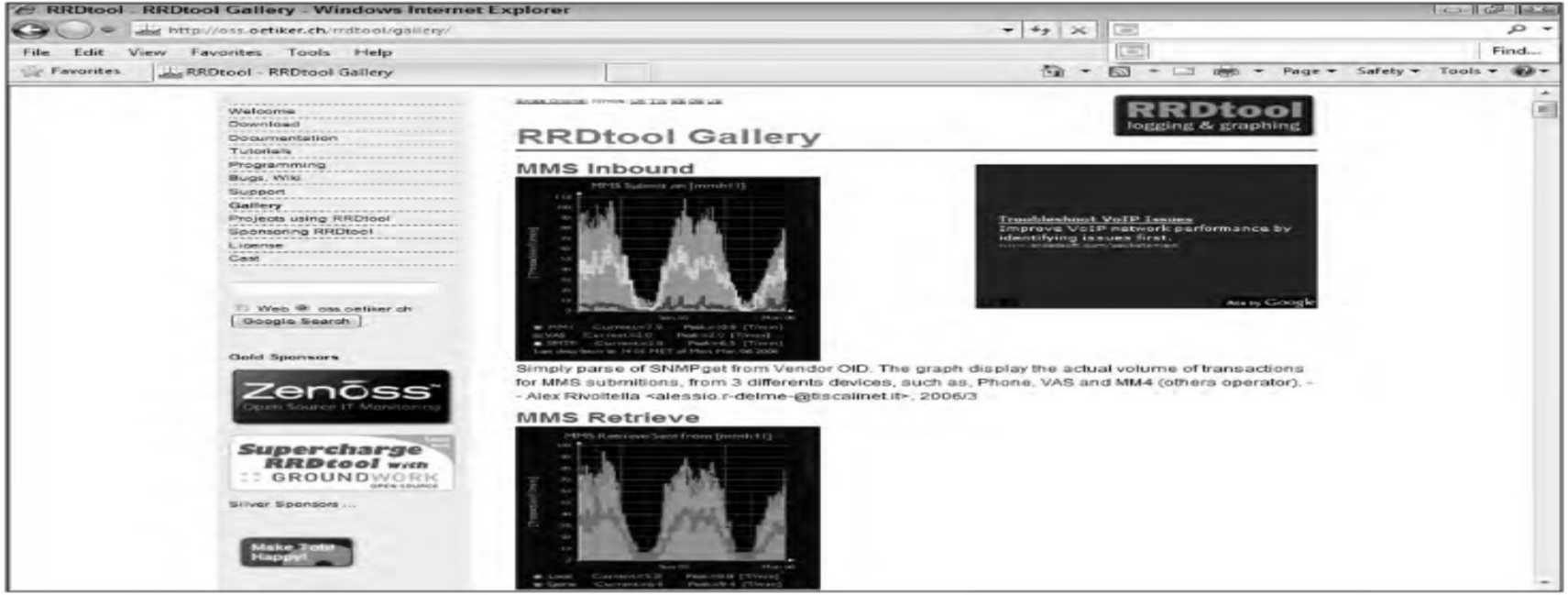
- Each of these resources can be measured by tools that are **operating-system-specific,** but for which tools that are their counterparts exist for all operating systems.
- Indeed, many of these tools were written specifically to build the operating systems themselves, but that's another story.
- In Linux/UNIX, you might use the sar command to display the level of CPU activity. sar is installed in Linux as part of the sysstat package.
- In Windows, a similar measurement may be made using the Task Manager, the data from which can be dumped to a performance log and/or graphed.
- Some tools give you a historical record of performance, which is particularly useful in capacity planning.
- For example, the popular Linux performance measurement tool **RRDTool(Round Robin Database tool) http://oss.oetiker.ch/rrdtool/)** is valuable in this regard.

System Metrics

- RRDTool is a utility that can **capture time-dependent performance data from resources such as a CPU load, network utilization (bandwidth),** and so on and store the data in a circular buffer.
- It is commonly used in **performance analysis work**. A time interval in RRDTool is called a step, with the value in a step referred to as a primary data point (PDP). When a function is applied to a data point.
- The function is referred to as a **Consolidation Function (CF),** and the value obtained is a **Consolidated Data Point (CDP)**. An interval is stored in the **Round Robin Archive (RRA),** and when that interval is filled, it is replaced by new step data.
- RRDTool includes a **graphical front end for displaying performance results visually**. RRDTool is widely used for a number of different purposes.
- Gallery of RRDTool graphs found at **http://oss.oetiker.ch/rrdtool/gallery/.**

System Metrics

RRDTool lets you create historical graphs of a wide variety of performance data. Some sam
in the gallery at http://oss.oetiker.ch/rrdtool/gallery/.



lists some LAMP performance testing tools.

System Metrics

LAMP Performance Monitoring Tools

Tool Name	Web Site	Developer	Description
Alertra	http://www.alertra.com	Alertra	Web site monitoring service
Cacti	http://www.cacti.net	Cacti	Open source RRDTool graphing module
Collectd	http://www.collectd.org/	collectd	System statistics collection daemon
Dstat	http://dag.wieers.com/home-made/dstat/	DAG	System statistics utility; replaces vmstat, iostat, netstat, and ifstat
Ganglia	http://www.ganglia.info	Ganglia	Open source distributed monitoring system
Gomez	http://www.gomez.com	Gomez	Commercial third-party Web site performance monitor
GraphClick	http://www.arizona-software.ch/graphclick/	Arizona	A digitizer that can create a graph from an image
GroundWork	http://www.groundworkopensource.com/	Groundwork's Open Source	Network monitoring solution
Hyperic HQ	http://www.hyperic.com	Spring Source	Monitoring and alert package for virtualized environments
Keynote	http://www.keynote.com	Keynote	Commercial third-party Web site performance monitor
Monit	http://www.tildeslash.com/monit	Monit	Open source process manager
Munin	http://munin.projects.linpro.no/	Munin	Open source network resource monitoring tool
Nagios	http://www.nagios.org	Nagios	Metrics collection and event notification tool
OpenNMS	http://www.opennms.org	OpenNMS	Open source network management platform
Pingdom	http://www.pingdom.com	Pingdom	Uptime and performance monitor
RRDTool	http://www.RRDTool.org/	Oetiker+Partner AG	Graphing and performance metrics storage utility
SiteUpTime	http://www.siteuptime.com	SiteUpTime	Web site monitoring service
Zabbix	http://www.zabbix.com	Zabbix	Performance monitor
ZenOSS	http://www.zenoss.com/	Zenoss	Operations monitor, both open source and commercial versions

Load Testing

- Examining your server under load for system metrics isn't going to give you enough information to do meaningful capacity planning.
- **Load testing** is a type of performance testing where you **simulate real-world user activity to see how a system, application, or server performs under expected or peak loads.**
- You need to know what happens to a system when the load increases. Load testing seeks to answer the following questions:
- What is the maximum load that my current system can support?
- Which resource(s) represents the bottleneck in the current system that limits the system's performance?
- This parameter is referred to as the **resource ceiling.** Depending upon a server's configuration, any resource can have a bottleneck removed, and the resource ceiling then passes onto another resource.
- Can I alter the configuration of my server in order to increase capacity?
- How does this server's performance relate to your other servers that might have different characteristics

Load Testing

Steps in Load Testing :

- Define objectives – e.g., test if a web app can handle 1,000 users.
- Identify metrics – response time, throughput, CPU, memory usage.
- Create test scenarios – simulate typical user actions.
- Run the test – gradually increase the load.
- Analyze results – check for slowdowns, errors, or crashes.
- Optimize and retest – fix issues and repeat.

Type	Purpose
Normal Load	Test system under expected traffic.
Peak Load	Test system at maximum expected traffic.
Stress Load	Test beyond max to see where it fails.
Endurance	Test system over long periods to detect memory leaks.

Load Testing

- If you have one production system running in the cloud, then overloading that system until it breaks isn't going to make you popular with your colleagues.
- However, cloud computing offers virtual solutions. One possibility is to create a clone of your single system and then use a tool to feed that clone a recorded set of transactions that represent your application workload.
- Two examples of applications that can replay requests to Web servers are :
- **HTTPerf (http:// hpl.hp.com/research/linux/httperf/) and Siege (http://www.joedog.org/ JoeDog/Siege),**
- These tools run the requests from a single client, which can be a resource limitation of its own.
- You can **run Autobench (http://www.xenoclast.org/auto bench/)** to run HTTPerf from multiple clients against your Web server, which is a better test.

Load Testing

Consider these load generation tools as well:

- HP LodeRunner (https://h10078.www1.hp.com/cda/hpms/display/main/
- hpms_content.jsp?zn=bto&cp=1-11-126-17^8_4000_100__)
- IBM Rational Performance Tester (http://www-01.ibm.com/software/ awdtools/tester/performance/)
- JMeter (http://jakarta.apache.org/jmeter)
- OpenSTA (http://opensta.org/)
- Micro Focus (Borland) SilkPerfomer (http://www.borland.com/us/products/ silk/silkperformer/index.html)
- Load testing software is a large product category with software and hardware products.
- Load testing useful in testing the performance of not only Web servers, but also application servers, database servers, network throughput, load balancing servers, and applications that rely on client-side processing.

Load Testing

- When you have multiple virtual servers that are part of a server farm and have a load balancer in front of them, you can use your load balancer to test your servers' resource ceilings.
- This technique has the dual advantages of allowing you to slowly increment traffic on a server and to use real requests while doing so.
- Most load balancers allow you to weight the requests going to a specific server with the goal of serving more requests to more powerful systems and fewer requests to less powerful systems.
- Sometimes the load balancer does this optimization automatically, and other times you can exert manual control over the weighting.
- Whatever method you use to load a server for performance testing, you must pick a method that is truly representative of real events, requests, and operations. The best approach is to incrementally alter the load on a server with the current workload.
- When you make assumptions such as using a load balancer to serve traffic based on a condition that summarizes the traffic pattern instead of using the traffic itself, you can get yourself into trouble.

Load Testing

- Problems with load balancers have led to some spectacular system failures because those devices occupy a central controlling site in any infrastructure.

- For example, if you assume that traffic can be routed based on the number of connections in use per server and your traffic places a highly variable load based on individual requests, then your loading measurements can lead to dramatic failures when you attempt to alter your infrastructure to accommodate additional loads.

Resource ceilings :

- **Resource ceilings** refer to the **maximum limits set on system resources** that a process, user, or application is allowed to consume.

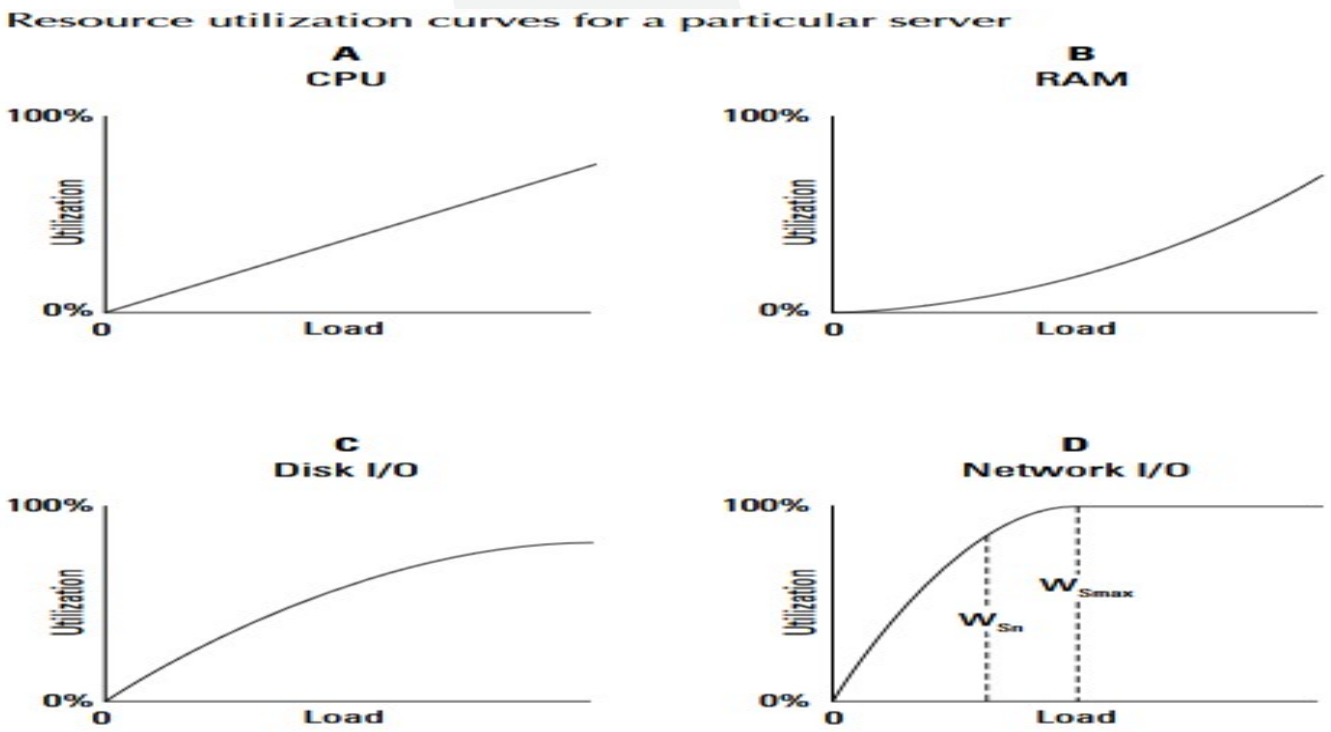
- Whatever performance measurement tool you use, **the goal is to create a set of resource utilization curves similar to the ones shown in Figure 6.3 for individual server types in your infrastructure.** To do this, you must examine the server at different load levels and measure utilization rates.

- The figure indicate that over a certain load for a particular server, **the CPU (A), RAM (B), and Disk I/O (C)** utilization rates rise but do not reach their resource ceiling. The **Network I/O (D)** reaches its maximum 100-percent utilization at about 50 percent of the tested load, and this factor is the current system resource ceiling.

Load Testing

- Network I/O is often a bottleneck in Web servers, and this is why, architecturally, Web sites prefer to scale out using many low-powered servers instead of scaling up with fewer but more powerful servers.
- Adding more (multi-homing) or faster network connections can be another solution to improving Web servers' performance.
- Unless you alter the performance of the server profiled in Figure 6.3 to improve it, you are looking at a maximum value for that server's workload of Wsmax as shown in the dashed line at the 50-percent load point in graph D.
- The server is overloaded and the system begins to fail. Some amount of failure may be tolerable in the short term, provided that the system can recover and not too many Web hits are lost, but this is a situation that you really want to minimize.
- Consider WSn to be the server's red line, the point at which the system should be generating alerts or initiating scripts to increase capacity.

Load Testing



Load Testing

- The parameter you are most interested in is likely to be the overall system capacity, which is the value of WT . **WT is the sum over all the Web servers** in your infrastructure: **WT = Σ(WSnP WSnV).**

- In this equation, **WSnP represents the workload of your physical server(s) and WSnV is the workload of the virtual servers (cloud-based server instances) of your infrastructure.**

- The amount of overhead you allow yourself should be dictated by the amount of time you require to react to the challenge and to your tolerance for risk.

- A capacity planner would define a value WT such that there is sufficient overhead remaining in the system to react to demand that is defined by a number greater than WMAX by bringing more resources on-line.

- For storage resources that tend to change slowly, a planner might set the red line level to be 85 percent of consumption of storage; for a Web server, that utilization percentage may be different. This setting would give you a 15-percent safety factor where to draw the red line.

- When you load test a system, you are applying a certain amount of overhead to the system from the load testing too a feature that is often called the “observer effect.

Load Testing

- Many load testers work by installing lightweight agents on the systems to be tested. Those agents themselves impact the performance you see; generally though, their impact is limited to a few percent.
- Additionally, in order to measure performance, you may be forced to turn on various performance counters. A performance counter is also an agent of sorts;
- The counter is running a routine that measures some aspect of performance such as queue length, file I/O rate, numbers of READs and WRITEs, and so forth.
- While these complications are second order effects, it's always good to keep this in mind and not over-interpret performance results.
- Before leaving the topic of resource ceilings, let's consider a slightly more complicated case that you might encounter in MySQL database systems. Database servers are known to exhibit resource ceilings for either their file caches or their Disk I/O performance.
- To build high-performance applications, many developers replicate their master MySQL database and create a number of slave MySQL databases. **All READ operations are performed on the slave MySQL databases, and all WRITE operations are performed on the master MySQL database.**

Load Testing

- A Master / Slave database system has two competing processes and the same server that are sharing a common resource: **READs and WRITEs to the master/slave databases, and replication traffic between the master database and all the slave databases.**

- The ratio of these transactions determines the WSn for the system, and the WSn you derive is highly dependent upon the system architecture.

- These types of servers reach failure when the amount of WRITE traffic in the form of INSERT, UPDATE, and DELETE operations to the master database overtakes the ability of the system to replicate data to the slave databases that are servicing SELECT (query/READ) operations.

- The more slave databases there are, the more actively the master database is being modified and the lower the WS for the master database is.

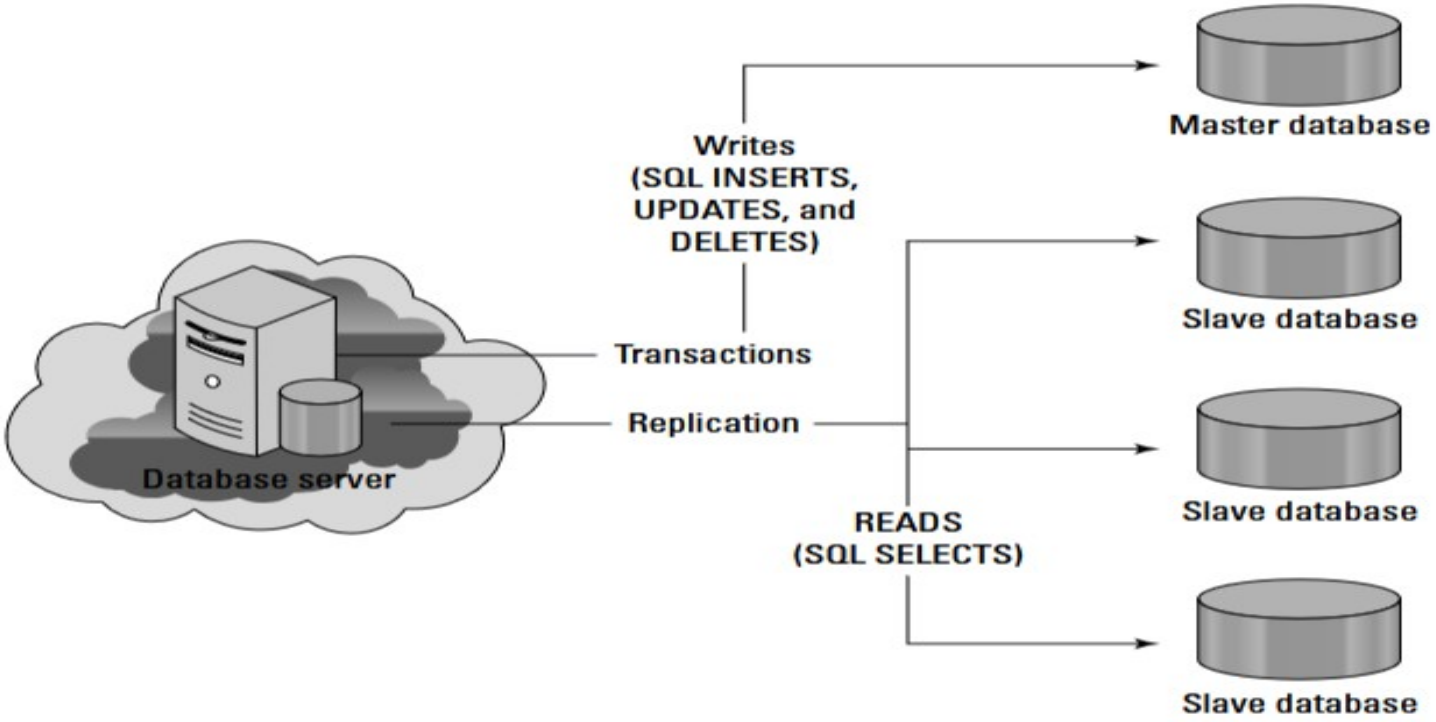
- This system may support no more than 25-35 percent transactional workload as part of its overall capacity, depending upon the nature of the database application you are running.

Load Testing

- Increase the working capacity of a database server that has a **Disk I/O resource ceiling** by using more powerful disk arrays and improving the interconnection or network connection used to connect the server to its disks.
- Disk I/O is particularly sensitive to the number of spindles in use, so having more disks equals greater performance. Keep in mind that your ability to alter the performance of disk assets in a virtual or cloud-based database server is generally limited.
- In the first case, a single server has a single application with a single resource ceiling. In the second case, you have a single server that has a single application with two competing processes that establish a resource ceiling.
- What do you do in a situation where your server runs the full LAMP stack and you have two or more applications running processes each of which has its own resource ceiling?
- In this instance, you must isolate each application and process and evaluate their characteristics separately while trying to hold the other applications' resource usage at a constant level.
- Do this by examining identical servers running the application individually or by creating a performance console that evaluates multiple factors all at the same time. Remember, real-world data and performance is always preferred over simulation

Load Testing

Resource contention in a database server



Load Testing

- As your infrastructure grows, it becomes more valuable to create a performance console that lets you evaluate your **KPIs (Key Performance Indicators)** graphically at an instant.
- Many **third-party tools offer performance consoles**, and some tools allow you to capture their state so you can restore it at a later point.
- An example of a performance analysis tool that lets you save its state is the **Microsoft Management Console (MMC).**
- In the Amazon Web Service, the statistics monitoring tool is called **Amazon CloudWatch**, and you can create a performance monitoring console from the statistics it collects.

Server and Instance Types

- **Server types** and **instance types** describe how **computing resources are organized and deployed in IT systems,** especially in data centers and cloud environments.

Server Types :

1. Physical Server (Bare Metal)
- Dedicated hardware
- High performance and full control
- Used for databases, legacy systems

2. Virtual Server
- Runs on a hypervisor
- Shares physical resources
- Flexible and cost-effective

3. Application Server
- Hosts business logic and applications
- Examples: Tomcat, WebLogic

4. Web Server
- Handles HTTP/HTTPS requests
- Examples: Apache, Nginx, IIS

5. Database Server
- Stores and manages data
- Examples: MySQL, PostgreSQL, Oracle

6. File Server
- Centralized file storage and sharing

Server and Instance Types

- One goal of capacity planning is to make growth and shrinkage of capacity predictable. Greatly improve your chances by standardizing on a few hardware types and then well characterizing those platforms.
- In a cloud infrastructure such as Amazon Web Server, the use of different machine instance sizes is an attempt to create standard servers.
- Reducing the variability between servers makes it easier to troubleshoot problems and simpler to deploy and configure new systems.
- As much as possible, you also should assign servers standardized roles and populate those servers with identical services.
- A server with the same set of software, system configuration, and hardware should perform similarly if given the same role in an infrastructure.

Server and Instance Types

- There is more performance variability in cloud machine instances than you might expect. This variability may be due to your machine instances or storage buckets being moved from one system to another system.
- Therefore, you should be attentive to the increase in potential for virtual system variability and build additional safety factors into your infrastructure. System instances in a cloud can fail, and it is up to the client to recognize these failures and react accordingly.
- Virtual machines in a cloud are something of a black box, and you should treat them as such because you have no idea what the underlying physical reality of the resources you are using represents.
- Capacity planning seeks to compare the capability of one system against another system and to choose the solution that is not only the right size but provides the service with the best operational parameters at the lowest cost.
- In Figure a graph of different server types is shown, with some hypothetical physical servers plotted against Amazon Machine Instance types.
- This type of graph allows you to add the appropriate server type to your infrastructure while performing a cost analysis of the deployment.

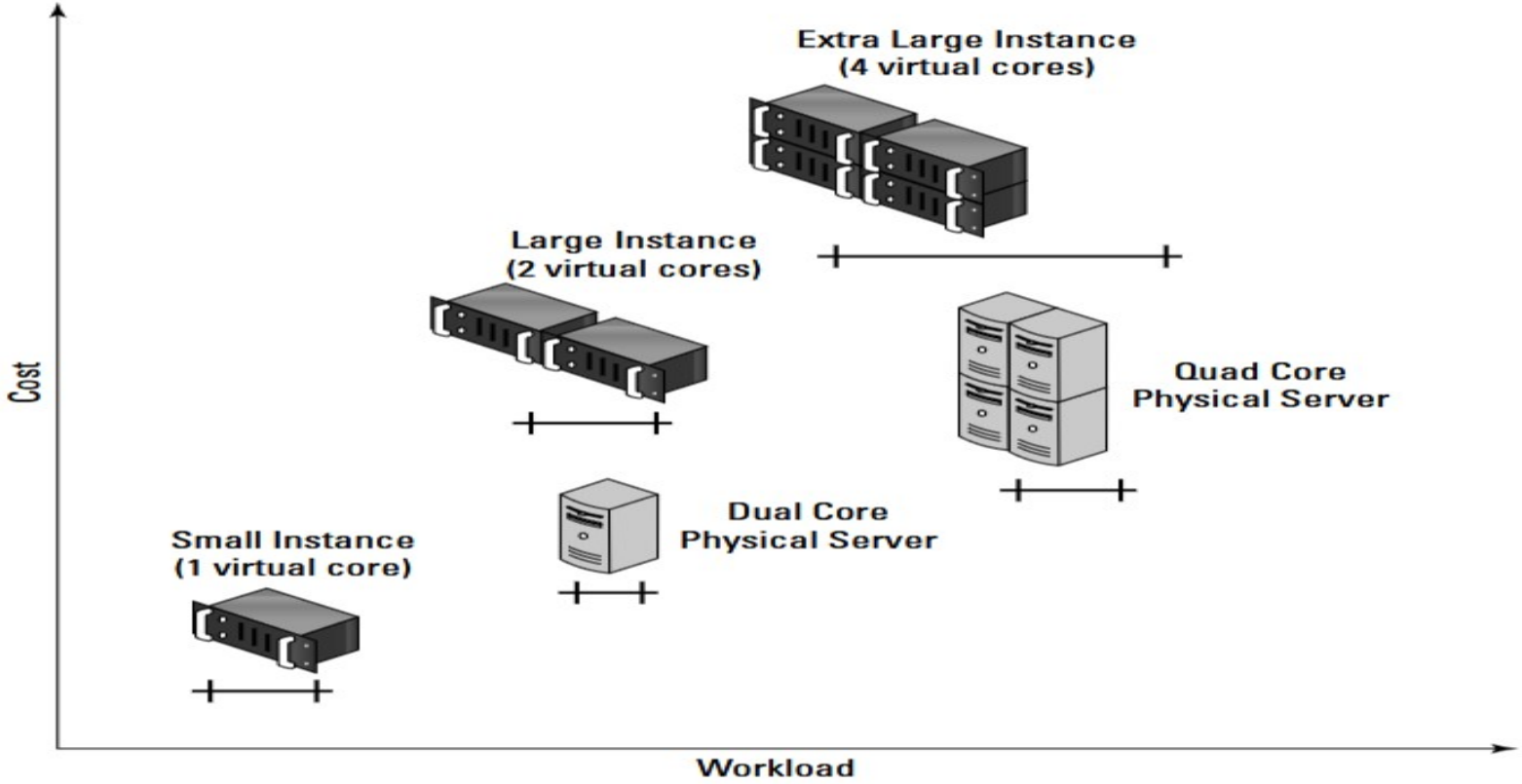
Server and Instance Types

An Amazon Machine Instance (AMI) is described as follows:

- **Micro Instance**: 633 MB memory, 1 to 2 EC2 Compute Units (1 virtual core, using 2 Cus for short periodic bursts) with either a 32-bit or 64-bit platform
- **Small Instance (Default):** 1.7GB memory, 1 EC2 Compute Unit (1 virtual core with 1 EC2 Compute Unit), 160GB instance storage (150GB plus 10GB root partition), 32-bit platform, I/O Performance: Moderate, and API name: m1.small
- **High-Memory Quadruple Extra Large Instance:** 68.4GB of memory, 26 EC2 Compute Units (8 virtual cores with 3.25 EC2 Compute Units each), 1,690GB of instance storage, 64-bit platform, I/O Performance: High, and API name: m2.4xlarge
- **High-CPU Extra Large Instance**: 7GB of memory, 20 EC2 Compute Units (8 virtual cores with 2.5 EC2 Compute Units each), 1,690GB of instance storage, 64-bit platform,
- **I/O Performance:** High, API name: c1.xlarge

Server and Instance Types

Relative costs and efficiencies of different physical and virtual servers



Server and Instance Types

- While you may not know what exactly EC2 Compute Unit or I/O Performance High means, at least we can measure these performance claims and attach real numbers to them.
- Amazon says that an EC2 Compute Unit is the equivalent of a 1.0-1.2 GHz 2007 Opteron or 2007 Xeon processor, but adds that "Over time, we may add or substitute measures that go into the definition of an EC2 Compute Unit, if we find metrics that will give you a clearer picture of compute capacity.
- What they are saying essentially is that this was the standard processor their fleet used as its baseline, and that over time more powerful systems may be swapped in.
- Whatever the reality of this situation, from the standpoint of capacity planning, all you should care about is measuring the current performance of systems and right-sizing your provisioning to suit your needs.
- In cloud computing, you can often increase capacity on demand quickly and efficiently.
- Not all cloud computing infrastructures automatically manage your capacity for you as Amazon Web Service's Auto Scaling (http://aws.amazon.com/autoscaling/) feature does.

Server and Instance Types

- For example, Amazon Web Services' infrastructure runs not only clients' machine instances, but also Amazon.com's machine instances as well as Amazon.com's partners' instances.
- If you need more virtual servers on Black Monday to service Web sales, you may be out of luck unless you have an SLA that guarantees you the additional capacity.
- Finding an SLA that you can rely on is still something of a crap shoot. If you are in the position of having to maintain the integrity of your data for legal reasons, this imposes additional problems and constraints.
- Easy to get started and requires little up-front costs. However, cloud computing is not cheap. At the point where you have a large cloud-based infrastructure, you may find that it is more expensive to have a metered per-use cost than to own your own systems.
- The long-term efficiencies are realized in the reduced need for IT staff and management, the ability to react quickly to business opportunities, and the freeing of a business from managing networked computer systems to concentrate on their core businesses.
- The Total Cost of Ownership (TCO) of a cloud-based infrastructure might be beneficial, it might be difficult to convince the people paying the bills that this is really so.

Network Capacity

Any cloud-computing system resource is difficult to plan for, it is network capacity. There are three aspects to assessing network capacity:

- **Network traffic to and from the network interface at the server,** be it a physical or virtual interface or server
- **Network traffic from the cloud to the network interface**
- **Network traffic from the cloud through your ISP to your local network interface (your computer)**
- This makes analysis complicated. You can measure factor 1, the network I/O at the server's interface with system utilities, as you would any other server resource.
- For a cloud-based virtual computer, the network interface may be a highly variable resource as the cloud vendor moves virtual systems around on physical systems or reconfigures its network pathways on the fly to accommodate demand.
- To measure network traffic at a server's network interface, you need to employ what is commonly known as a network monitor, which is a form of packet analyzer.
- Microsoft includes a utility called the Microsoft Network Monitor as part of its server utilities,

Network Capacity

The site Sectools.org has a list of packet sniffers at http://sectools.org/ sniffers.html.

- Wireshark (http://www.wireshark.org/), formerly called Ethereal
- Kismet (http://www.kismetwireless.net/), a WiFi sniffer
- TCPdump (http://www.tcpdump.org/)
- Dsniff (http://www.monkey.org/~dugsong/dsniff/)
- Ntop (http://www.ntop.org/)
- EtherApe (http://etherape.sourceforge.net/)

Network Capacity

Factor 2 is the cloud's network performance, which is a measurement of WAN traffic. A WAN's capacity is a function of many factors:

- Overall system traffic (competing services)
- Routing and switching protocols
- Traffic types (transfer protocols)
- Network interconnect technologies (wiring)
- The amount of bandwidth that the cloud vendor purchased from an Internet backbone provider Again, factor 2 is highly variable and unlike factor 1, it isn't easy to measure in a reliable way.
- Tools are available that can monitor a cloud network's performance at geographical different points and over different third-party ISP connections.
- This is done by establishing measurement systems at various well-chosen network hubs.

Network Capacity

- A company that makes WAN network monitoring software, has set up a series of these points of presence at various Internet hubs and uses its networking monitoring software called PathView Cloud to collect data in a display that it calls the Cloud Performance Scorecard (http://www.apparentnetworks.com/CPC/scorecard.aspx).
- Figure shows this Web page populated with statistics for some of the cloud vendors that Apparent Networks monitors.
- Use PathView Cloud as a hosted service to evaluate your own cloud application's network performance at these various points of presence and to create your own scorecard of a cloud network.
- Current pricing for the service is $5 per network path per month. The company also sells a small appliance that you can insert at locations of your choice and with which you can perform your own network monitoring.
- The last factor, factor 3, is the connection from the backbone through your ISP to your local system, a.k.a.
- Internet connection is more like an intelligently managed thin straw that you are desperately trying to suck information out of. So Factor 3 is measurable, even if the result of the measurement isn't very encouraging, especially to your wallet.

Network Capacity

- Internet connectivity over the last mile (to the home) is the Achilles heel of cloud computing. The scarcity of high-speed broadband connections (particularly in the United States) and high pricing are major impediments to the growth of cloud computing.

- Many organizations and communities will wait on the sidelines before embracing cloud computing until faster broadband becomes available.

- Indeed, this may be the final barrier to cloud computing's dominance. That's one of the reasons that large cloud providers like Google are interested in building their own infrastructure, in promoting high-speed broadband connectivity, and in demonstrating the potential of high-speed WANs.

- Google is running a demonstration project called Google Fibre for Communities **(http://www.google.com/appserve/fiberrfi/public/overview)** that will deliver 1 gigabit-per-second fiber to the home.

- The few lucky municipalities chosen (with 50,000 to 500,000 residents) for the demonstration project will get advanced broadband applications, many of which will be cloud-based.

Network Capacity

Apparent Networks' Cloud Performance Center provides data on WAN throughput and uptime at Internet network hubs.

Scaling

- In capacity planning, after you have made the decision that you need more resources, you are faced with the fundamental choice of scaling your systems.
- You can either scale vertically (scale up) or scale horizontally (scale out), and each method is broadly suitable for different types of applications.
- To scale vertically, you add resources to a system to make it more powerful. For example, during scaling up, you might replace a node in a cloud-based system that has a dual-processor machine instance equivalence with a quad-processor machine instance equivalence.
- You also can scale up when you add more memory, more network throughput, and other resources to a single node. Scaling out indefinitely eventually leads you to an architecture with a single powerful supercomputer.
- Vertical scaling allows you to use a virtual system to run more virtual machines (operating system instance), run more daemons on the same machine instance, or take advantage of more RAM (memory) and faster compute times.
- Applications that benefit from being scaled up vertically include those applications that are processor-limited such as rendering or memory-limited such as certain database operations—queries against an in-memory index.

Scaling

- **Horizontal scaling** or scale out adds capacity to a system by adding more individual nodes.
- In a system where you have a dual-processor machine instance, you would scale out by adding more dual-processor machines instances or some other type of commodity system.
- Scaling out indefinitely leads you to an architecture with a large number of servers (a server farm), which is the model that many cloud and grid computer networks use.
- Horizontal scaling allows you to run distributed applications more efficiently and is effective in using hardware more efficiently because it is both easier to pool resources and to partition them.
- Although your intuition might lead you to believe otherwise, the world's most powerful computers are currently built using clusters of computers aggregated using high speed interconnect technologies such as InfiniBand or Myrinet.
- Scale out is most effective when you have an I/O resource ceiling and you can eliminate the communications bottleneck by adding more channels.Web server connections are a classic example of this situation.

Scaling

Aspect	Horizontal Scaling	Vertical Scaling
Method	Add more servers	Upgrade one server
Cost	Incremental	Can be expensive
Fault tolerance	High	Low
Scalability	High	Limited
Complexity	Higher	Lower

- **Horizontal scaling** and **vertical scaling** are two approaches used to increase a system's capacity and performance.
- **Vertical Scaling** increasing the **capacity of a single server** by adding more CPU, RAM, or storage.

Scaling

- These broad generalizations between scale up and scale out are useful from a conceptual stand point, but the reality is that there are always tradeoffs between choosing one method for scaling your cloud computing system versus the other.

- The pricing model that cloud computing vendors now offer their clients isn't fully mature at the moment, and you may find yourself paying much more for a high-memory extra-large machine instance than you might pay for the equivalent amount of processing power purchased with smaller system equivalents.

- This has always been true when you purchase physical servers, and it is still true when purchasing virtual servers. Cost is one factor to pay particular attention to, but there are other tradeoffs as well.

- Scale out increases the number of systems you must manage, increases the amount of communication between systems that is going on, and introduces additional latency to your system.